CSCE 46103/56103 Artificial Intelligence – Fall 2026
Intelligent Agents and Task Environments

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Right Decision, Wrong Outcome?

Scenario

An AI coding agent examines the specification, repository, and visible tests. It chooses a patch that passes every visible test.

Was that choice rational?

Available evidence
specification,
files, visible tests

↓

Chosen patch
chosen using
that evidence

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Visible outcome
all visible tests pass

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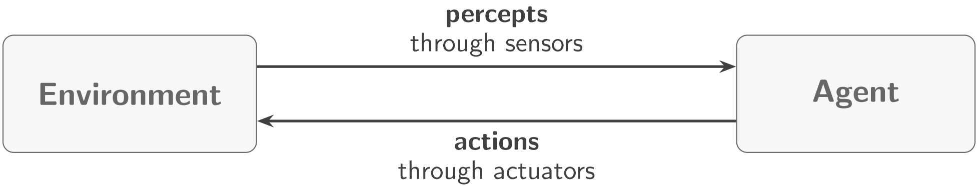
Later Evaluation

A hidden test exposes an edge-case failure in the patch.
Does that prove irrationality? What else must we know?

The Perception–Action Loop



Concept: Russell and Norvig [2020, Sec. 2.1].

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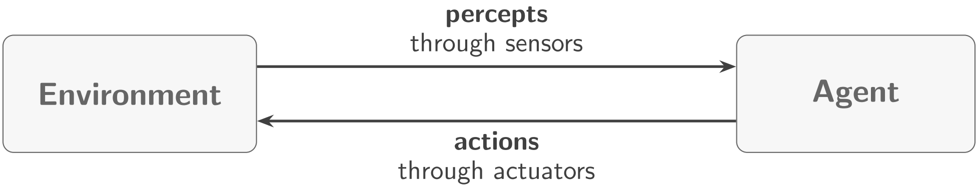


AIMA's Analytical View

An agent perceives an environment through sensors and acts on it through actuators.

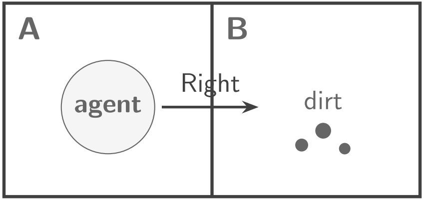
Coding environment
Repository, runtime, tests, user

Percepts
Files, tool results, messages

Actions
Edits, commands, questions, replies

Concept: Russell and Norvig [2020, Sec. 2.1].

Agent Functions and Percept Histories



Vacuum World

Percepts report the agent's location and whether its current square is dirty. Actions are `Left`, `Right`, `Suck`, and `NoOp`.

Vacuum-world example and terminology: Russell and Norvig [2020, Sec. 2.1].

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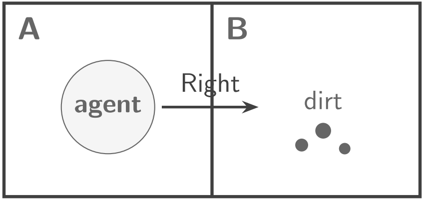

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Percept and Percept Sequence

A **percept** is one observation, such as $[A, \text{Dirty}]$. The **percept sequence** is the complete observation history.

$$f : \mathcal{P}^* \to \mathcal{A}$$

percept histories $\longrightarrow$ actions

Function and Program

The **agent function** describes behavior abstractly. The **agent program** implements that mapping on an architecture.

Vacuum-world example and terminology: Russell and Norvig [2020, Sec. 2.1].

Rationality: Evidence and Objectives

Rational Action

A performance measure evaluates any sequence of environment states. For each possible percept sequence, a rational agent selects an action expected to maximize that measure, given the percept evidence and its prior knowledge.

Rationality depends on: measure | prior knowledge | available actions | percept history

Not Hindsight

Rational does not mean knowing the future, achieving the best actual outcome, or succeeding every time.

A Bad Measure Rewards Bad Behavior

"Visible tests pass" can reward weakening tests. The intended criterion is software that is correct, within scope, and supported by verification evidence.

Information gathering, exploration, and learning can themselves be rational.

Rationality, information gathering, learning, and autonomy: Russell and Norvig [2020, Sec. 2.2].

PEAS for an Automated Taxi

Automated taxi

P — Performance
Safety, legality, travel time, comfort, cost

E — Environment
Roads, traffic, pedestrians, passengers, weather

A — Actuators
Steering, throttle, brakes, signals, display

S — Sensors
Cameras, lidar, GPS/maps, speedometer, passenger input

PEAS and automated-taxi example: Russell and Norvig [2020, Sec. 2.3.1].

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Design principle: Define PEAS before designing the agent—start with the performance measure.

PEAS and automated-taxi example: Russell and Norvig [2020, Sec. 2.3.1].

PEAS for the Coding Agent

PEAS element	Coding-agent task specification
Performance	Specification correctness; preservation of valid tests; security; maintainability; scoped changes; clear explanation; verification evidence
Environment	Repository, dependencies, test runner, runtime, user, and permitted external services
Actuators	Edit files; run approved commands and tests; ask questions; report changes and evidence
Sensors	Specification, files, diagnostics, test output, diffs, and tool or user feedback

Two Important Distinctions

Visible test output is **percept evidence**, not the objective. Functional correctness is part of the **performance criterion**. The tool interface determines which sensors and actuators actually exist.

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Observability and Outcome Models

What Can the Agent Sense?

Fully observable: sensors reveal all action-relevant state.
Partially observable: evidence is incomplete or noisy.
Unobservable: no relevant state information is sensed.

What Can Follow an Action?

Deterministic: state and action(s) fix the next state.
Nondeterministic: several outcomes are possible.
Stochastic: probabilities over outcomes are supplied.

Do Not Conflate Them

Partial observability hides the current state; nondeterminism allows multiple successors even when the current state is known.

Observability and outcome models: Russell and Norvig [2020, Sec. 2.3.2].

Knowledge and Observability

Two Different Questions

Observable: Can the agent sense the relevant current state?
Known: Does it know the action model and relevant probabilities?

Action model	Fully observable	Partially observable
Known model	Chess: board state visible; action rules known	Solitaire: rules known; face-down cards hidden
Unknown model	New video game: full state visible; button effects unknown	Unfamiliar building: limited robot sensors and unknown action effects

Knowledge of the model and observability of the current state are independent axes.

Known/unknown and observable/partially observable: Russell and Norvig [2020, Sec. 2.3.2].

Temporal and Multiagent Dimensions

Dimension	Diagnostic question	Automated taxi
Episodic / Sequential	Are decisions independent, or can one decision affect what follows?	Sequential
Static / Dynamic / Semidynamic	During a decision, does the world stay fixed, change, or only impose a time-dependent score?	
Discrete / Continuous	Are state, time, percepts, and actions distinct or continuously varying?	
Single-Agent / Multiagent	Do other agents pursue interacting objectives?	

Classify the automated taxi task.

Task-environment dimensions: Russell and Norvig [2020, Sec. 2.3.2].

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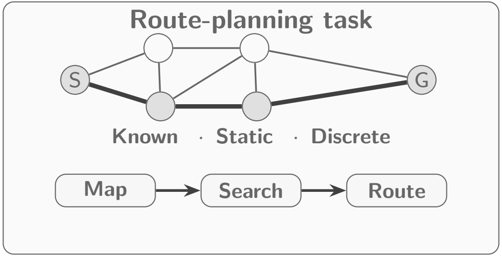
Single-Agent / Multiagent	Do other agents pursue interacting objectives?	Multiagent

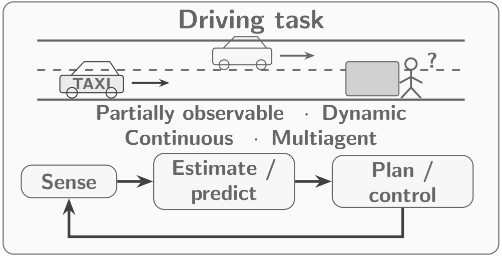
The classification depends on the task boundary and level of detail.

Task-environment dimensions: Russell and Norvig [2020, Sec. 2.3.2].

Why Task Classification Matters

Can a route planner drive the taxi?





Method–Task Mismatch

The route can be correct while the next steering or braking action is unsafe.

Classification exposes the assumptions a method may safely make.

Design implications based on Russell and Norvig [2020, Chs. 2–4].

Task-Environment Classification

1

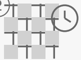
Two-room vacuum

Ex.	Observability	Agents	Outcome	Temporal	Change	Values	Model
1	Partially observable	Single-agent	Deterministic	Sequential	Static	Discrete	Known

Classification dimensions and examples: Russell and Norvig [2020, Sec. 2.3.2].

Task-Environment Classification


1 **Two-room vacuum**


2 **Chess + clock**

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Task-Environment Classification

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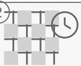
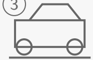
2  **Chess + clock**

3  **Taxi**

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Task-Environment Classification

1. **Two-room vacuum**
2. **Chess + clock**
3. **Taxi**
4. **Coding repair**

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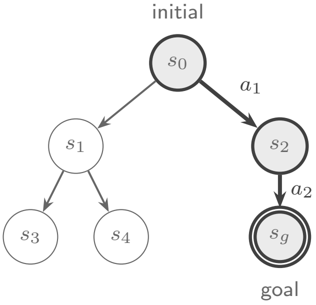
Classification follows the task boundary, not the application label.

Classification dimensions and examples: Russell and Norvig [2020, Sec. 2.3.2].

Classical Search

State-space view



Thick path: returned plan

What Is Classical Search?

Starting from a known initial state, classical search uses a known action model to generate successors and explores the resulting state space for an action sequence that reaches a goal, often while minimizing path cost.

Initial state	Where the search starts
Actions + result	Available choices and their successors
Goal test	Whether a state satisfies the objective
Path cost	Cost of the complete action sequence

Classical problem-solving agents and search: Russell and Norvig [2020, Secs. 3.1–3.3].

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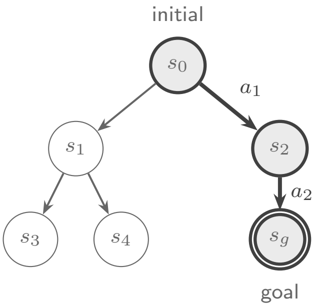

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Typical starting assumptions

- **Fully observable**
- **Single-agent**
- **Deterministic**
- **Sequential**
- **Static**
- **Discrete**
- **Known model**

Classical problem-solving agents and search: Russell and Norvig [2020, Secs. 3.1–3.3].

Opening Question Revisited

Can the right decision still fail?

Rationality and the problem-solving-agent transition: Russell and Norvig [2020, Secs. 2.2 and 3.1].

Opening Question Revisited

Can the right decision still fail?

Yes—Outcome Alone Does Not Determine Rationality

A hidden failure does not prove irrationality. Judge the choice by expected performance given the intended objective and information available at the time.

Rational but Unlucky

The expected-performance-maximizing choice can still produce a bad outcome.

Irrational but Lucky

A poorly justified choice can produce a good outcome by chance.

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Bridge to Problem Formulation

Full taxi driving
partial observation
uncertain, dynamic
multiagent

→

Route planning
observable state
known action model
deterministic

→

Problem formulation
state space, initial state
actions and results
goals and costs

Rationality and the problem-solving-agent transition: Russell and Norvig [2020, Secs. 2.2 and 3.1].

Thank You

Any questions?

References

S. J. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2020. URL https://aima.cs.berkeley.edu/.

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